

Study 5G NR Architecture

Radio Access Network: The Radio Access Network provides a wireless connection between the user devices and 5G core Network.

- 1) **gNB:** The gNB is the 5G base station that provides radio connectivity between the User Equipment and the 5G core.

Functions:

- Provides wireless communication
- Allocates radio resources
- Performs beamforming
- Handles handovers between neighboring cells
- Manages scheduling and radio transmissions

The gNB is equivalent to the 4G eNodeB but offers significantly improved performance and supports advanced 5G features.

- 2) **UE (User Equipment):** User Equipment (UE) refers to any device that connects to the 5G network.

Examples:

- Smartphones
- Tablets
- Laptops
- IoT devices
- Smart Sensors

Functions:

- Connects to the 5G network
- Sends and receives data
- Performs authentication with the network

- Maintains mobility while moving between cells
- Requests internet and other network services

5G Core Network: The 5G Core (5GC) is responsible for authentication, session management, mobility, and data forwarding.

1) **AMF (Access and Mobility Management Function):** The AMF manages signaling and mobility for users.

Responsibilities:

- User registration
- Authentication
- Security management
- Mobility management
- Connection management

Whenever a mobile device powers on or moves between cells, the AMF manages the signaling procedures.

2) **SMF (Session Management Function):** The SMF manages data sessions.

Responsibilities:

- Creates PDU sessions
- Assigns IP addresses
- Controls Quality of Service (QoS)
- Selects the appropriate UPF
- Manages user connectivity

The SMF acts as the session controller of the network

3) **UPF (User Plane Function):** The UPF handles the forwarding of user data packets.

Responsibilities:

- Packet routing
- Traffic forwarding
- QoS enforcement
- Traffic filtering
- Internet connectivity

The UPF is responsible for carrying the user's actual internet traffic.

Interfaces in 5G: Different network functions communicate using standardized interfaces.

Interface	Connected Components	Purpose
N1	UE ↔ AMF	Registration, authentication, NAS signaling
N2	gNB ↔ AMF	Control plane signaling and mobility management
N3	gNB ↔ UPF	User data transfer
N4	SMF ↔ UPF	Controls user sessions and forwarding rules
N6	UPF ↔ Data Network	Internet and external network connectivity

5G Architecture Block Diagram:

Data Network

(Internet/Cloud)

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N6

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UPF

User Plane Function

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N4

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SMF

Session Management

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AMF

Access & Mobility

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N2

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GNB

5G Base Station

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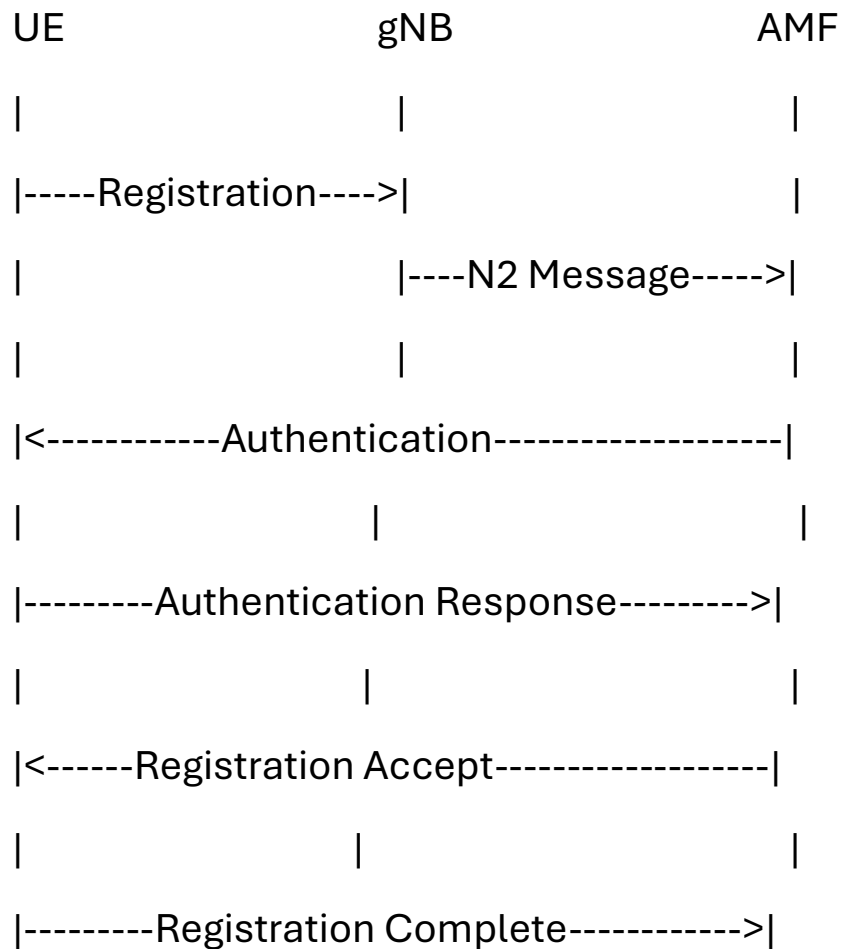
N1

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UE

Smartphone/IoT

5G Registration Call Flow:



Data Session Flow:

UE

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| User Data

gNB

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AMF

|

|

SMF

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|

UPF

|

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Internet/Data Network